How Gender Attributes Play a Role in User Identification

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***Abstract*—Our research aims to understand the effect of gender with facial recognition through authentication, with regards to a Support Vector Machine classification system and a K Nearest Neighbors Classification system. Current societal characteristics amongst women and men can be exhibited through facial features such as facial hair among men and makeup among women. By separating the CelebA image dataset into images pertaining to gender specific facial obstructions, we were able to conduct findings on the accuracies of multi-feature specific classifications with gender identity at the forefront. Score level fusion is used to combine the results of Support Vector Machine and K-Nearest Neighbors facial classification. Based on the results of our accuracy levels, male facial recognition had a margin of higher accuracy compared to women by 3%, which aligns with findings from other research papers [1][2][3][4][7]. Furthermore, female facial recognition with and without heavy makeup had an accuracy change of 3%, with heavy makeup and lipstick accounting for an accuracy boost. The accuracy differences between genders as well as obstruction subcategories suggest that gendered obstruction may not play a well-defined role in substantial accuracy changes. However, it comes to a surprise that identities with gender specific obstructions seemed to perform better than identities without obstructions. This could be due to previous research, where facial obstructions such as facial hair have led to a greater averaged distance of separation between identities [3].**

Keywords—facial recognition, gender bias, biometrics.

I. Introduction

Authentication is deeply engrained into the fabric of society. Digital security has become increasingly important as more and more sensitive information is stored online. Biometrics are the measurements and statistical analysis of people's unique physical and behavioral characteristics. This can be any trait innate to a person like their face, fingerprint, or even gait. Biometric systems leverage these traits to authenticate individuals, taking advantage of various types of sensors to capture biometric information and query the system against previously collected templates. One biometric system which has been put into widespread use is facial recognition. However, this method of authentication is not perfect and can be influenced by bias present in training and development. In this study, the influence of gender was observed on the accuracy of the facial authentication system we designed, specifically the influence of gendered facial obstructions. Attributes we considered to be gendered facial obstructions were facial hair for males and heavy makeup and lipstick for females. This is not exclusive, as there are women that can have facial hair and men that wear makeup. But for the purposes of this paper, we have made a general assumption about the population and the dataset we worked with.

The purpose of this paper was to explore two research questions: how does gender affect the accuracy of facial recognition? What is the role of male and female based facial attributes in facial authentication with respect to accuracy?

First, all the data was concatenated together into one txt file, comprising all the image attributes, facial landmarks, identities, and image file names. To make the dataset manageable for the model and to balance the number of images used for each identity, a subset is filtered to only contain identities that have a certain number of images. This ensures that no identities are unequally represented and would be better recognized than other identities. Furthermore, the number of identities between men and women are equal, and the four subgroups of male or female obstruction/non-obstruction were approximately balanced using a majority decision labeling algorithm (see methods for more information). The result of this new subset is converted to three npy files, one with landmarks, one with corresponding identities, and another with the matching attributes, in order to make classification more efficient.

The Euclidean distance between the landmarks of each images are taken to create a feature vector which represents the distance between each image. These feature vectors were then split into training and test sets and used to train an SVM classifier and a KNN classifier. These classifiers would test training set identities against the dataset, giving the identity a true claimed identity if it was supposed to represent a genuine score and a random other identity if it was supposed to be an imposter score. Once the model was confirmed to work, we

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filtered the test set to only include images with certain attributes. We tested males and females, then males with facial hair and females with heavy makeup, then males without facial hair and females with no makeup. There were some small differences in accuracy found between these subsets, but the differences are small enough to potentially be unrelated to gender and facial obstructions.

II. Related Work

Outside research has found that women typically experience lower facial recognition accuracy than men [1][2][3][4][7]. However, it is unclear what the general underlying cause of this discrepancy is. One research paper aims to observe gendered differences in face recognition based on hairstyles, makeup, and facial morphology. They found significant differences between the head sizes and shapes of women and men as well as differences in how prominent the ears and sides of the face are. This is because women tend to have hairstyles which obscure the sides of the face whereas men tend to have shorter hair which allows for the ears and sides of the face to be considered for facial recognitions. Because of this, there are typically fewer "pixels on the face" for women than men, making it so that images of women have less information provided than men [1]. They also observed the impact of makeup on the accuracy of facial recognition by balancing the number of people wearing makeup in both the male and female datasets. This improved the discrepancy in false negative rates between men and women [1]. Finally, they tested face clusters to observe why the false positive rates for women tended to be higher. They found women tended to have closer facial similarities from woman to woman than men did, which could account for the gender disparity in face recognition [1].

Another study done found similar results, finding that men typically have hairstyles that contribute to a greater average difference between male faces [3]. Additionally, women's hairstyles tended to occlude more of the face, resulting in less information being provided to the facial recognition model. When the datasets were balanced in terms of facial occlusions, the gender gap in accuracy largely disappeared [3].

Another consideration is the balance of men and women in a dataset. Some people theorize that because women tend to be less represented in datasets, that is why their accuracy score tends to be smaller. However, this was found to be untrue by a separate study which compared the accuracy scores for men and women with datasets that had different percentages of male and female images. When the datasets were balanced there was still a discrepancy, which was only diminished with a 25% male to 75% female ratio [2]. However, while this does solve the disparity, this maximizes accuracy for neither men nor women and is thus inefficient [2]. The report gives substantial reasoning that even with highly tuned datasets, female facial authentication still underperforms over male facial authentication, meaning another alternative theory is cause for this issue.

III. Dataset

CelebA is the dataset we will be using to conduct our research. This dataset contains:

- 10,177 number of identities
- 202,599 number of face images
- 5 landmark locations
- 40 binary attributes annotations per image [9]

IV. Features

The CelebA image dataset contains a variety of feature labels and positional landmarks for the 200,000+ images. For our research purposes, we will be focusing on the following features and landmarks:

5-positional Landmarks: lefteye_x lefteye_y righteye_x righteye_y nose_x nose_y leftmouth_x leftmouth_y rightmouth_x rightmouth_y

Each landmark contains an (x,y) coordinate position based on the aligned image.

Gender Specific Features: heavy makeup, lipstick, gender, mustache, beard, goatee, and 5' o clock shadow. These are attributes typically associated with either gender.

Each feature has a binary value of 1 or –1. 1 means that the feature is present. -1 means that the feature is not present. [9]

V. Methods

The CelebA dataset has been repurposed for the objective of this report. We start with **img_align_celeba,** a folder containing 200,000+ facial images that are normalized (cropped and aligned). These images have corresponding feature labels, identities, and facial landmarks all separated into different .txt pages. To simplify the process of data separation, function **labelConcatenation()** creates one *combinedLabels.txt,* which is a combination of all the provided information that can be paired with a given image. Combining information will be helpful when creating sub .txt files for image separation.

It is critical that through our evaluation of male and female facial verification, we must ensure a balanced dataset is provided to our classifiers. **Balanced_Dataset()** produces the balanced dataset where the following criteria are met:

1. The data set should contain an n number of images for each identity.
2. The data set should contain an equal number of men and women identities.
3. The data set should contain an approximated similar number of images with and without obstructions based on gender.

An identity is deemed to be imbalanced if its number of images are not equal to its corresponding other identities. Function **identityNumRange()** allowed us to identify the number of identities in CelebA that have the same n number of images. For our classification, we only choose identities

who have exactly 20 images, ensuring balance between all identities with respect to data size.
To avoid the issue that "How Does Gender Balance In Training Data Affect Face Recognition Accuracy?" [2] showcases with gender imbalances in datasets, we further refine our data set to ensure the number of male identities equal female identities.

There is no right or wrong way to properly execute the third criteria. Given a male identity where the identity shares 30 images, 10 of their images contain facial hair, while 20 contain no facial hair. To ensure that a balance of non-obstructions/obstructions are represented during training, we use an approximate balancing by sampling identities approach. Function **classify_identity()** uses a majority decision approach, where it observes an identity, its gender, and its corresponding images, and classifies an identity as an obstruction case or non-obstruction case based on the higher amount. In our male identity example, this identity will be paired with a non-obstruction label, since 20 > 10, or we have a majority of non-obstruction compared to the total image count. Figure 1. showcases our resulting identity labeling.

TABLE I. IDENTITY LABELLING

Male Identities With Obstruction Majority	Male Identities Without Obstruction Majority	Female Identities With Obstruction Majority	Female Identities Without Obstruction Majority
428	144	365	107

Fig. 1. The resulting group seperation from classify_identity() when image count = 20

Once every identity collected has a corresponding obstruction/non-obstruction label, we find the value of an identity group that has the least amount of their corresponding label (in our case, 107) and normalize the rest of the labels to match that amount.

Following all three of these filtration procedures, we end up with an identity set of **428 identities**, where 214 identities are male and the other 214 female.

To extract the features present in the dataset, the function **extract_features()** gets all unique pairs of landmarks and finds the Euclidean distances between them. The generated features are then split into a testing and training set. For the purposes of addressing the research questions, images with certain attributes were retrieved and a portion of those were used as queries. If our system was to be used for authentication, this would not need to be done. Then both models, the Support Vector Machine (SVM) and K Nearest Neighbor (KNN) models would be trained on the training sets. A number is randomly generated to determine if a query will be a genuine or imposter query. If it is a genuine query, the claimed identity will be its true identity, and the **y_hat** array will put a 1 in the corresponding index of the array. For an imposter query, the claimed identity will be a random different identity, and the **y_hat** array will have a 0 at that index. Then the SVM and KNN models will be fitted to the training set and asked to determine the probability of the query being an imposter or genuine score. **Score level fusion** is performed on the probabilities given by the SVM and KNN classifiers. This way, each classifier can compensate for the potential flaws of the other. The probabilities of the genuine scores are added to the **genuine_scores** array and the probabilities of the imposter scores are in the **imposter_scores** array. These are used to generate the ROC and DET curves. The predictions are tracked in the **y_pred** array, and the accuracy score is found by comparing the array of predicted values to the array of actual values.

VI. RESULTS

The authentication system was able to achieve an overall accuracy of about 75%. Then the accuracies were calculated for each subset of genders. The ROC and DET curves were also generated for each gender. These show the overall performance of the authentication system. They all indicate some level of overlap between the genuine and imposter scores which makes it difficult to adequately separate and distinguish them from each other.

TABLE II. ACCURACY BASED ON GENDER AND GENDERED FACIAL OBSTRUCTIONS.

Accuracy		
Gender	***Obstruction***	***Accuracy***
Male	All obstructions included	0.76
Female	All obstructions included	0.73
Male	Facial Hair	0.75
Male	No Facial Hair	0.74
Female	Heavy Makeup and Lipstick	0.74
Female	No Makeup or Liptick	0.71

Fig. 2. Accuracy scores based gender groups and gendered facial obstructions.

As shown in Figure 2, accuracy for men was higher than for women, which is in line with the findings of prior studies. This is potentially because the beards defined a more unique face shape for men which made them easily distinguishable. However, because the accuracy scores are so close, it is hard to say this is the case conclusively.

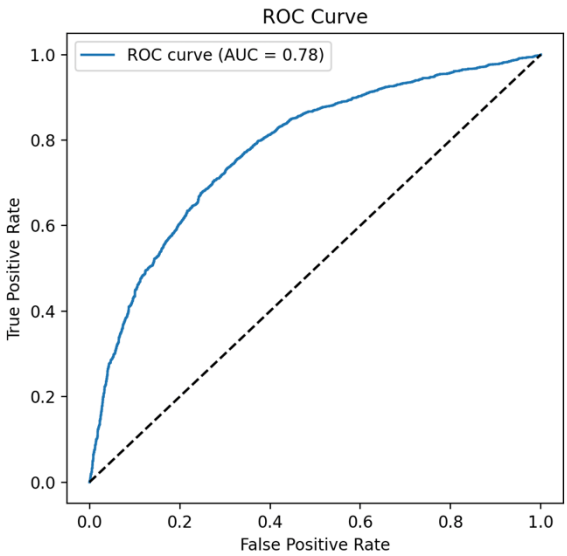

Fig. 3. ROC curve for authentification for just males.

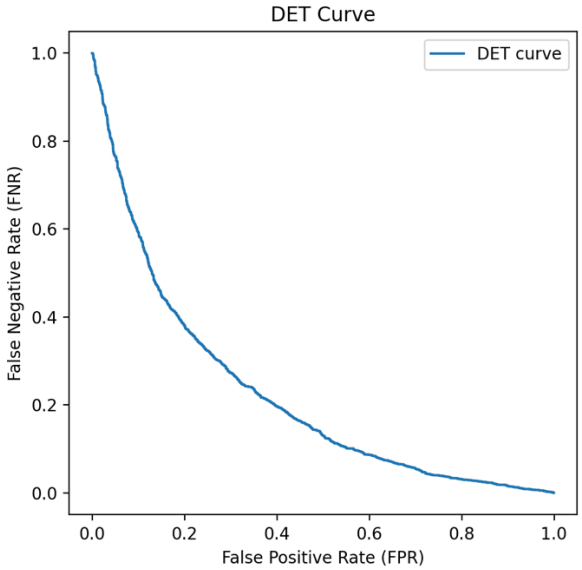

Fig. 4. DET curve for authentication for just men.

The area under the curve (AUC) of the ROC function indicates the combined models' ability to distinguish between different classes, in this case the many identities tested. The high AUC score is a good indicator that the authenticator is working as expected and could potentially be further improved in the future.

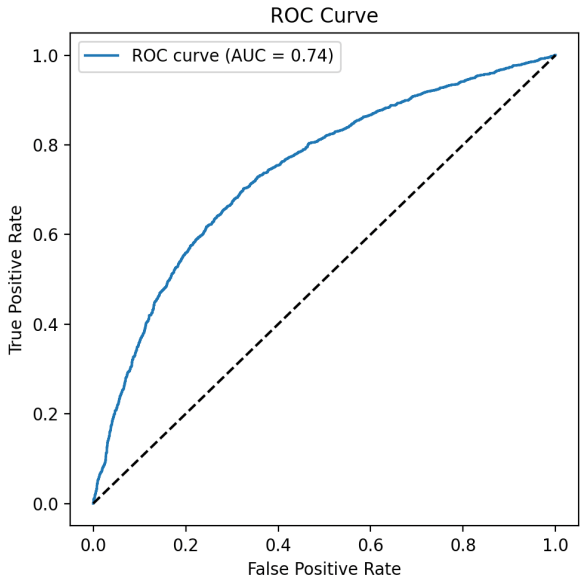

Fig. 5. ROC curve for authentication for just women.

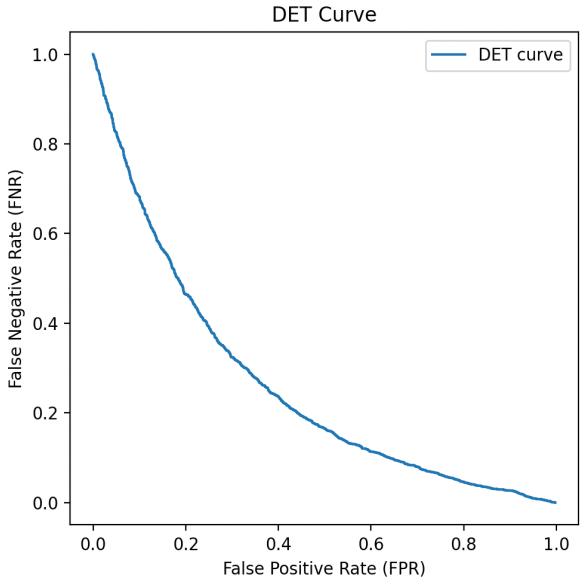

Fig. 6. DET curve for authentication for just women.

The ROC and DET curves for women are not as defined at the men's. Their AUC values are only 0.74 in comparison to the men's 0.78 score, indicating that the model is not as able to distinguish between female identities. This could be due to women having more similar facial structure than men.

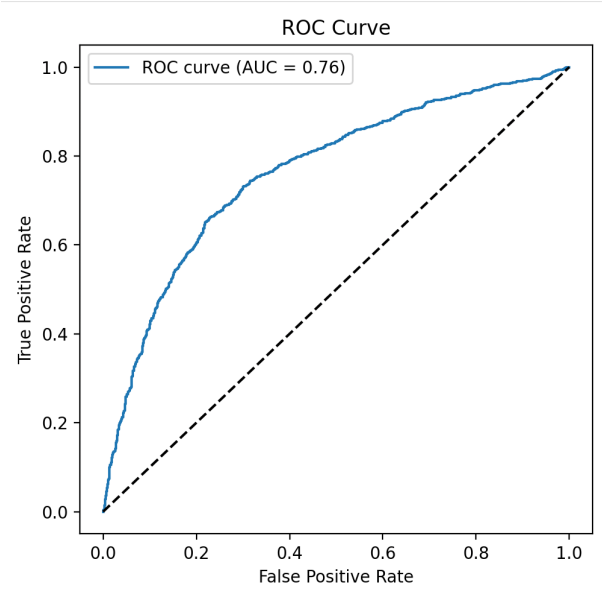

Fig. 7. ROC curve for authentication for men with facial hair.

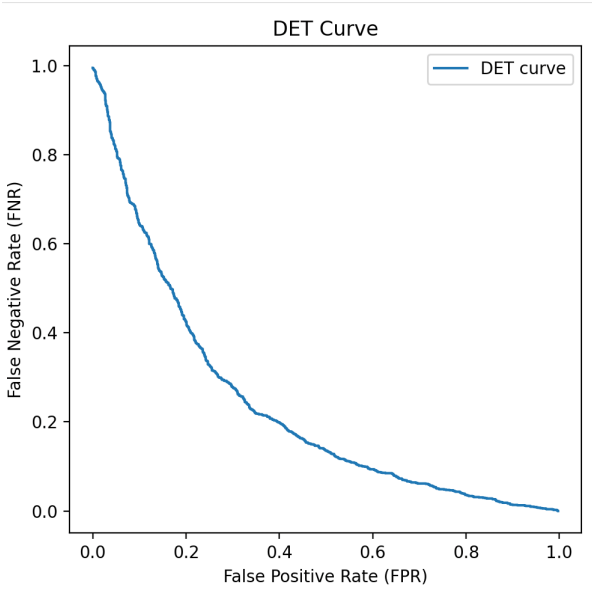

Fig. 8. DET curve for authentication for men with facial hair.

The DET and ROC curves for men with facial hair are better for men with facial hair than they are for women overall. Even though facial hair might typically be considered to be a facial obstruction, the accuracy was higher than men with no facial hair. This could be due to facial hair giving a unique shape to men's head shapes, making them easier to distinguish.

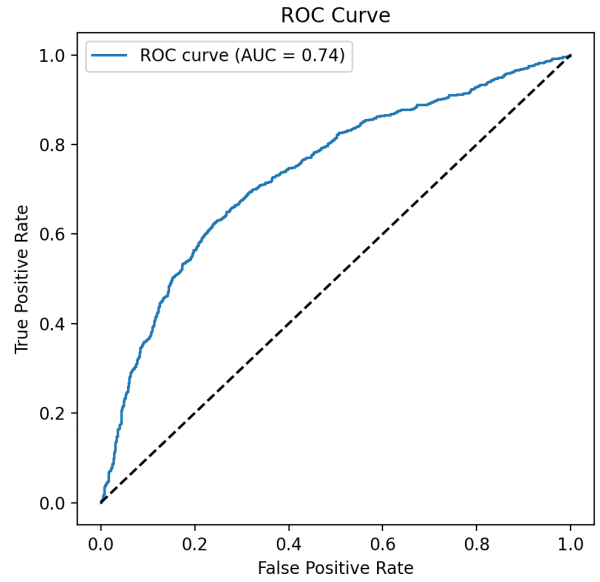

Fig. 9. ROC curve for authentication for men with no facial hair.

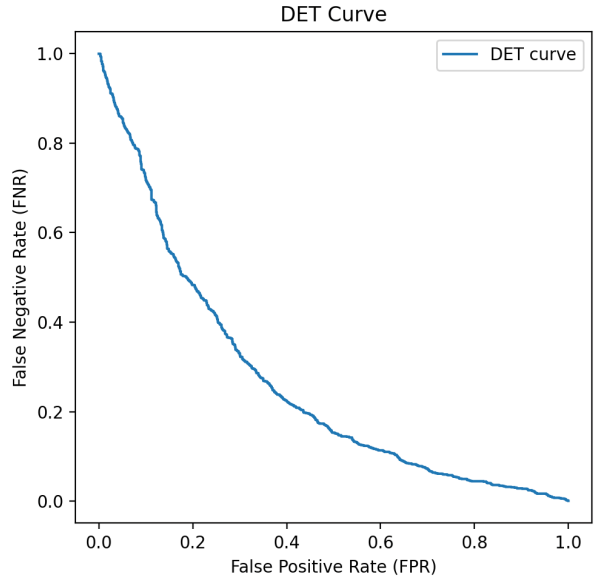

Fig. 10. DET curve for authentication for men with no facial hair.

The AUC score and the overall curve seem similar to the ones of the women's accuracy. This could indicate that a general lack of facial hair might make face shape less distinct. However, the accuracy score of men with no facial hair is only 0.75 in comparison to the accuracy of women at 0.74. This difference may not be large enough to be significant.

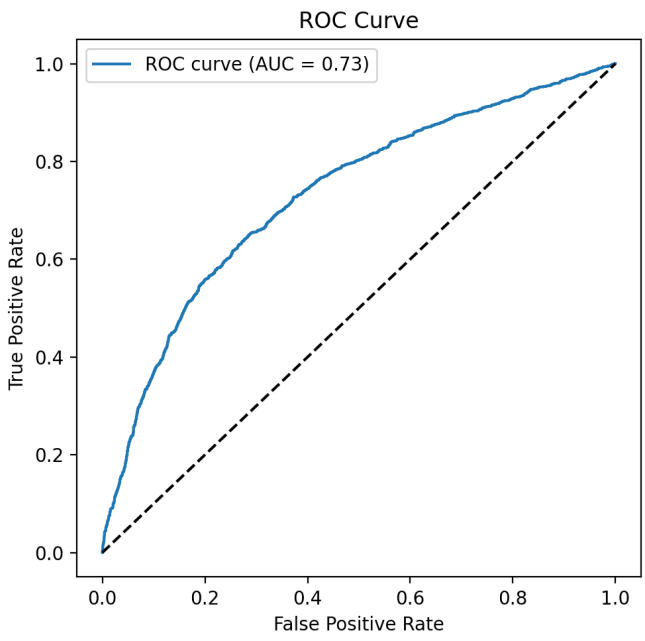

Fig. 11. ROC curve for authentication for women with heavy makeup.

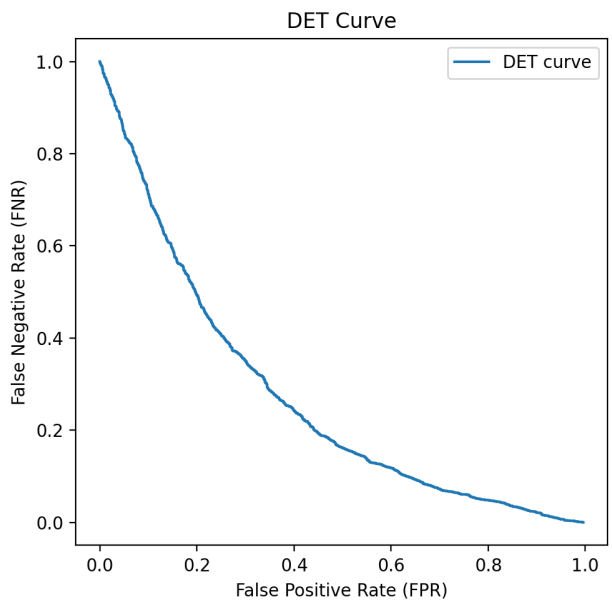

Fig. 12. DET curve for authentication for women with heavy makeup.

The ROC and DET curves for women with heavy makeup were surprisingly similar to the overall curves for women. We originally considered heavy makeup to be a potentially obstruction, but because many of the women in the dataset were celebrities, most of their images had heavy makeup and the dataset was trained on those images as a result.

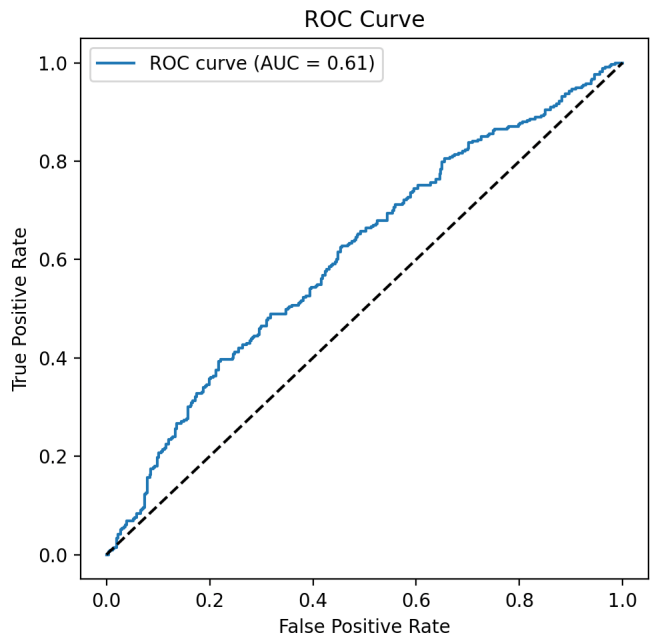

Fig. 13. ROC curve for authentication for women without heavy makeup.

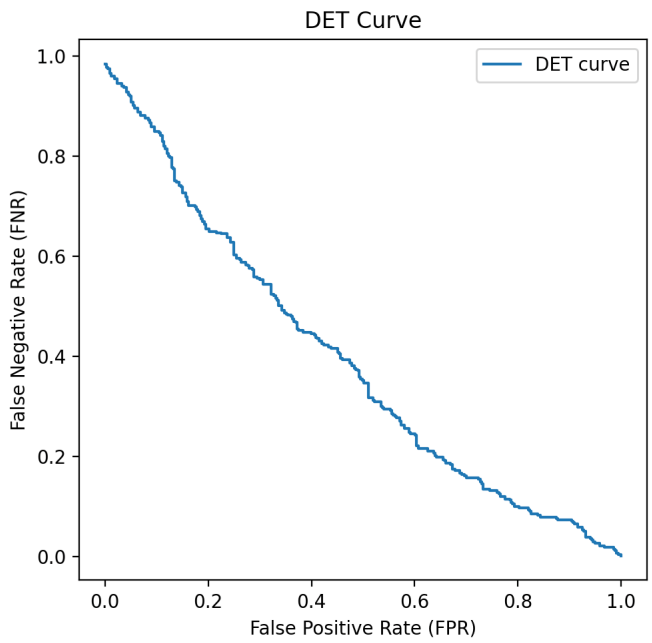

Fig. 13. DET curve for authentication for women without heavy makeup.

The ROC and DET curves for women with no makeup is by far the worse out of all the subsets. This could be due to the limited number of images which had no makeup or lipstick present.

Additionally, the score level fusion using the KNN model and the SVM classifier slightly improved the system accuracy, as the SVM classifier mainly focuses on creating clear margins

between identities and the KNN model compares their similarities.

VII. Ethics Statement

Potential Harms to Individuals

The primary ethical concerns in developing our facial recognition system pertain to privacy and security risks for individuals using the system. Facial recognition inherently involves sensitive biometric data, which, if improperly handled, could lead to breaches of personal privacy or unauthorized access to secure systems. Specific to our research, the CelebA dataset includes gendered facial features, which may carry the risk of biased outcomes if improperly managed. To protect individual privacy, we anonymized all identities in our dataset and ensured no personally identifiable information (PII) was used. Moreover, all data used in this study was publicly available and consented for research purposes. In a deployed system, obtaining explicit informed consent from users and implementing robust encryption protocols to protect stored biometric templates would be mandatory to minimize privacy risks.

Potential Negative Societal Impacts

Our system's potential for misuse must also be addressed. Facial recognition technology, particularly when biased or inaccurately trained, can increase societal inequities, such as gender discrimination or racial profiling. For example, our findings on gendered facial obstructions suggest that biases in classification accuracy could lead to unfair treatment of certain groups if integrated into systems like surveillance or hiring platforms. Additionally, misuse of facial recognition by authoritarian governments or in other unethical contexts could infringe on individuals' rights to privacy and autonomy. Awareness of these risks highlights the importance of careful consideration of the deployment context and regulation.

Risk-Mitigation Strategies and Citations

To mitigate these risks, we propose several strategies: first, continuous auditing of the system for bias using diverse, representative datasets; second, complying with ethical AI principles such as fairness, transparency, and accountability; and third, strict data protection measures, including encryption and secure storage protocols. Implementing privacy protection techniques and explaining system decisions to users could also increase trust and reduce potential harm. Finally, policy recommendations supporting transparent governance of biometric systems could minimize societal misuse. These strategies align with insights from existing literature on AI ethics and biometric security, including works by Leslie et al. (2020) on protecting civil liberties in AI [5], Roundtree (2022) on FRT Code of Ethics [8], and Jain et al. (2004) on biometrics privacy [10].

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